Course Syllabus

STAT 109: Introductory Biostatistics — Summer Session 1, 2026 (online, async)

STAT 109: Introductory Biostatistics

Cal Poly Humboldt — Summer Session 1, 2026

Course Information

Course Number: STAT 109
Course Title: Introductory Biostatistics
Units: 4
Session: Summer Session 1, 2026 (June 1 – July 2, 2026)
Mode of Instruction: Online, asynchronous
General Education: Lower Division GE Area B4 – Mathematical Concepts & Quantitative Reasoning

How this course works

- **Course website** (home): readings, labs, project instructions, reference sheets, and practice quiz banks.
- **Canvas**: submit work, take graded quizzes, see feedback and grades, and join module discussions.

There are no required live meetings. Work through each module on your own schedule and I'll provide support by answering questions, giving feedback and providing encouragement as you encounter obstacles.

Teacher: Dr. Rosanna Overholser
Email: rho3@humboldt.edu
Office Location: BSS 334
Office Phone: 707-826-4022
Office Hours: In an Canvas discussion and by appointment (email to schedule). I will respond to email within one business day during the session.

Please use your Cal Poly Humboldt email account to contact me and check both Canvas and your Cal Poly Humboldt email regularly for course communications.

Course Description (Catalog Description)

Descriptive statistics, probability, random variables, discrete and continuous distributions, confidence intervals, contingency tests, regression and correlation, tests of hypothesis, analysis of variance. Emphasis: methods and applications used in the biological and natural resource sciences.

Division: Lower Division · **Instruction Mode:** Online (async)

Prerequisite: MATH 101 or MATH 101I or MATH 102 or MATH 109B (may be taken concurrently with instructor approval) with a grade of C- or better or ALEKS PPL assessment score over 65 or appropriate high school coursework. See Math Placement and ALEKS PPL for details. MATH 102 may be taken concurrently with instructor approval.

Note: if you plan to use STAT 109 as a prerequisite for another MATH, DATA or STAT course, you must have a grade of C- or better in this course.

Course Learning Outcomes

By the end of this course, students will be able to:

1. Use numerical and graphical descriptive statistics to summarize and interpret data.
2. Explain the role of randomness, probability, and sampling in statistical inference.
3. Apply probability rules and work with common probability distributions.
4. Construct and interpret confidence intervals for means and proportions.
5. Conduct and interpret basic hypothesis tests.
6. Analyze relationships between pairs of variables using correlation and regression.
7. Use a statistical programming language to analyze data and communicate results.
8. Distinguish between correlation and causation and recognize limitations of statistical conclusions.

General Education & Institutional Learning Outcomes

This course fulfills Lower Division General Education Area B4 (Mathematical Concepts & Quantitative Reasoning) and supports Cal Poly Humboldt's Institutional Learning Outcomes related to quantitative reasoning, critical thinking, and communication.

Course Materials & Workload

This is a five-week intensive version of STAT 109. You should expect to spend approximately 30-40 hours per week on readings, reference-sheet updates, labs, quizzes, discussions, and projects.

Required Materials

- **Course website:** Course path, lectures, methods map, R reference, resources.
- **Software:** R in **Google Colab** (free; no local install required). Lab and project notebooks link to Colab from the course site. Optional: install R and RStudio on your own computer (resources).
- **Reference sheets:** You will build and update two personal references over the term.
- **Calculator:** A basic scientific calculator is helpful for by-hand checks; most computation is in R.
- **Computer Access**: you will need a computer with internet access and a web browser.

Optional Materials

Textbook: Introductory Statistics for the Life and Biomedical Sciences by Julie Vu and Dave Harrington, available for free online.

Access & Affordability

Students who do not have personal access to required technology may use campus computer labs and library resources. If you want to get a free used computer, please check whether our campus's Reuseable Office Supply Exchange (ROSE) has one available.

Fees

There are no additional course fees beyond standard tuition and registration fees.

Course Feedback & Assessment

Component	Weight	Description
Class Participation	5%	Class Chatter discussions in Canvas, one per Module Q+A Discussion
Quizzes	25%	Low-stakes practice of key terms, concepts and code syntax from readings and labs Best of 4 attempts

Component	Weight	Description
R Coding	20%	Read Rosanna's code, then try writing your own.
Exercise Labs		Graded on completion
Projects 1–5	50%	Personalized projects for your portfolio, highlighting your ability to apply the course material to meaningful questions.
Total	**100%**	

Practice quiz banks on the course site are open-ended. Canvas quizzes may draw a random subset and are autograded where possible so you'll have received immediate feedback.

Projects

Five portfolio projects apply course methods to meaningful questions. Each includes a one-page summary.

Project	Module	Due (2026)
Project 1 — Testing a Claim	1	Mon **Jun 8**
Project 2 — Estimating the Truth	2	Mon **Jun 15**
Project 3 — Discovering Patterns	3	Mon **Jun 22**
Project 4 — Designing Fair Comparisons	4	Mon **Jun 29**
Project 5 — Looking Ahead	5	Thu **Jul 2**

Discussions

One community "Class Chatter" discussion per module (Modules 0–5). These build connection in an async course; they are graded on participation, not statistical correctness.

Exams

There are no midterm or final exams in this async session. Exam-style items are integrated into synthesis quizzes and project work instead.

Grading

Final letter grades will be assigned based on overall performance:

A: 93–100% · A-: 90–92% · B+: 87–89% · B: 83–86% · B-: 80–82% · C+: 77–79% · C: 73–76% · C-: 70–72% · D: 60–69% · F: 0–59%

Note:

- **Grade Mode:** Letter grade by default. Students may elect Credit/No Credit grading according to university deadlines and procedures.
- A minimum grade of **C-** is required for the course to count toward GE Area B4.
- If you stop submitting work without communicating with me, you may receive a grade of WU (Withdrawal Unsatisfactory). Please contact me early if you are having difficulties.

Late or Missed Work Policies

All due dates in Canvas are just recommendations for pacing yourself through the material. If you want to turn in any assignment after it's recommended date, you may do so without penalty. All work that counts towards your course grade must be turned in by 11:59pm PST on July 2, 2026 in Canvas.

AI Policy

You may use AI tools (e.g. ChatGPT, NotebookLM, Copilot) to learn course material, explore code, or draft text, especially on take-home labs and projects.

For work you **submit for a grade**:

1. If you used AI, **acknowledge it briefly** (name of the tool is enough; you do not need to paste prompts).
2. You are **responsible** for the correctness, appropriateness, and academic integrity of everything you submit — AI-assisted or not.

I may use AI to improve assignment wording or check clarity of my teaching materials. I will not use AI to grade your work or run AI-detection tools on submissions.

Before the course started, I used AI to convert my teaching materials into this website with both html and pdfs versions in order to make my digital materials more accessible. Currently I believe only the equations on the html versions are fully accessible to screen readers so this is still a work in progress. However, having AI to help with this goal has been very helpful as previously my materials were scattered between various file types, some of which were not accessible at all for people using screen readers.

I also used the deep research function of ChatGPT to get an overview of the best teaching practices for online, asynchronous courses and the results of this informed my structuring of the material.

Course Topics & Schedule

Topics (by module)

Module	Theme	Core statistical ideas
0	Start here	Course tools, R in Colab, reference sheets
1	Testing a Claim	Probability, binomial process, hypothesis test with p-value
2	Estimating the Truth	Sampling, bias, confidence intervals
3	Discovering Patterns	Descriptive statistics, graphs, tidyverse
4	Designing Fair Comparisons	Study design, confounding, two-group inference
5	Looking Ahead	Correlation, simple linear regression, prediction

Important dates (Summer 2026)

This asynchronous version of STAT 109 is self-paced and the dates listed below are only recommendations. This is an intensive course, in which you are going through 4 semester units of material in only 5 weeks so you'll need to keep up. I'll be watching your process through the modules in Canvas and will do my best to support your completion with a minimum of stress.

Date	Event
Jun 2	Complete Module 0
Jun 2 – 8	Module 1 · Project 1
Jun 9 – 15	Module 2 · Project 2
Jun 16 – 22	Module 3 · Project 3
Jun 23 – 29	Module 4 · Project 4
Jun 30 – Jul 2	Module 5 · Project 5

University Policies & Resources

Students are responsible for reviewing the Cal Poly Humboldt Syllabus Addendum, which includes policies and resources related to:

- Academic honesty
- Disability accommodations
- Emergency procedures
- Title IX
- Student conduct and support services

Accommodations

Students who wish to request disability-related accommodations should contact the Student Disability Resource Center (SDRC) as early as possible and then email me (Rosanna Overholser, rho3@humboldt.edu) your letter of accommodation.

Online Conduct

Standards of respectful and professional behavior apply to all course interactions, including Canvas discussions, assignments, and email communication.

Changes to the Syllabus

This syllabus is subject to change. Any changes will be communicated through **Canvas announcements**.
